AI POWERED SMART ORCHID CARE SYSTEM USING MULTI-MODAL MACHINE LEARNING



Proposal Evaluation Feedback and Improvements

Reviewer Comments / Feedback	Actions Taken / Solutions Implemented
The presentation should be faster due to the limited presentation time.	We practiced and trained well to deliver the presentation faster and manage the limited presentation time effectively.
The proposal should clearly identify the number of nurseries/planters, project area coverage, number of plants considered for the cost calculation, and whether IoT devices are used per plant or per group of plants.	One farm contains approximately 5 houses with around 1800 plants. Each house uses 2 ESP32 cameras (One Hosue 2 cameras - approximately 10 cameras for the farm). IoT devices are used per group of plants, with 1 IoT device for approximately every 100 plants. Cost calculation per farm: • IoT modules: Rs. 45,000 • Cameras: 10 × Rs. 2,500 = Rs. 25,000 • Total cost: Approximately Rs. 70,000
The system should consider predicting diseases before symptoms become visible on the plant.	• Extended the scope to support early disease risk indication using environmental sensor data • The system alerts growers when environmental conditions increase the chance of disease development

AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

Problem & Impact:

Vanda orchid cultivation is difficult due to diseases, improper watering/fertilization, and unpredictable blooming. Manual monitoring can cause plant losses, reduced flower quality, and higher maintenance costs.

Who Will Benefit from the Solution

- Orchid farmers and commercial nurseries
- Home orchid growers and hobbyists
- Orchid breeders and researchers
- Agricultural institutions and greenhouse operators
- Floriculture businesses and export industries

Research Objective & Sub-Objectives

- Multi-modal Machine Learning techniques using image data, environmental sensor data, and historical growth patterns
- IoT-enabled smart monitoring for real-time environmental tracking
- Automated watering and fertigation systems for efficient resource management
- AI-based disease detection, severity assessment, and early prediction capabilities
- Growth stage classification and bloom prediction specifically designed for Vanda orchids
- Hybrid pollination compatibility analysis to support orchid breeding decisions
- Scalable and cost-effective architecture where shared IoT devices can monitor multiple plants





IT22065230 | Wickramasinghe D.P

Title of the Individual Component:
Hybrid pollination compatibility Analysis

Specialization: Software Engineering

The Solution Addresses the Sub-Problem/ Prototype

The prototype supports the first stage of hybrid pollination compatibility analysis by checking whether an orchid plant or flower is suitable to be used as a parent for pollination. Based on breeder feedback, the system considers visible traits such as leaf health, plant strength, disease visibility, and flower condition. It uses OpenCV-based feature extraction and machine learning to classify the orchid as Suitable, Moderate, or Not Suitable, helping growers make an initial pollination readiness decision before full parent-pair compatibility prediction.

User Requirements Addressed

- Upload orchid plant or flower image
- Select visible traits such as leaf condition, plant strength, and disease status
- Assess whether selected orchids are suitable for pollination
- Receive suitability result with confidence score
- Get guidance before selecting parent plants for hybrid pollination

Design Excellence / Contribution

- End-to-end ML pipeline from image upload to suitability prediction
- Real orchid plant and flower images used for feature extraction
- OpenCV features combined with visible plant traits for prediction
- Backend API integrated with the mobile application
- Feature importance supports breeder-based parent selection knowledge

User Feedback on Prototype

- Growers prefer simple inputs instead of technical terms
- Suitability results should be easy to understand
- Confidence score and recommendation text are useful for decision-making
- Users suggested adding direct comparison between two parent plants
- Future improvement: generate full parent-pair compatibility score







IT22064936 | Rathnamalala D.M.J.P.

Title of the Individual Component:
Orchid Disease Detection, Severity Assessment & Treatment Recommendation for Vanda Species

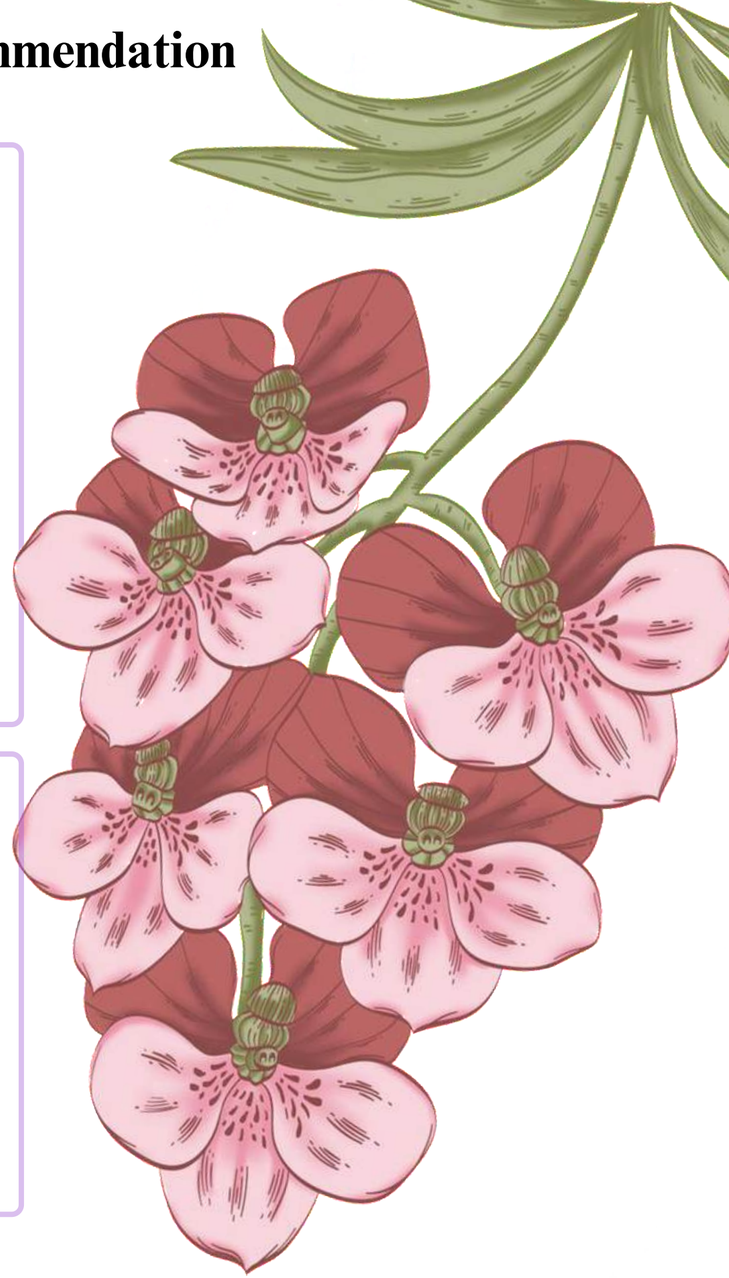
Specialization: Software Engineering

The Solution Addresses the Sub-Problem/ Prototype

The prototype uses AI-based image analysis to identify diseases in Vanda orchids using leaf, stem, and flower images. The system detects Phytophthora, Anthracnose, and Fusarium diseases while also distinguishing sunburn from actual diseases. This helps growers identify orchid health problems early and reduce manual monitoring.

User Requirements Addressed

- Automatic detection of orchid diseases from images
- Identification of multiple Vanda orchid disease types
- Differentiation between diseases and sunburn conditions
- Early detection to reduce disease spread
- Easy and efficient disease monitoring for orchid growers



Design Excellence / Contribution

- AI-powered disease detection for Vanda orchids
- Multi-modal approach using image and environmental sensor data
- Detects both diseases and non-disease conditions such as sunburn
- Improves early detection and decision-making accuracy
- Designed as a scalable smart disease management solution

User Feedback on Prototype

- Users found the system useful for identifying visible orchid diseases
- Disease detection can reduce manual monitoring effort and time
- Sunburn differentiation was considered valuable for practical use
- Users suggested improving accuracy with larger datasets and more disease samples







IT22062642 | Vithanage P. V. P. M

Title of the Individual Component:
Smart Watering and Automated Fertilization System for Vanda Orchids

Specialization: Software Engineering

The Solution Addresses the Sub-Problem/ Prototype

- ESP32 microcontroller reads soil moisture, temperature, humidity, and light intensity every 5 minutes and streams data to Firebase in real time
- Random Forest model analyzes 8 sensor features to predict whether watering is needed
- Decision Tree model determines fertilization requirements and recommends the correct NPK fertilizer type based on the plant's growth stage
- MobileNetV2 CNN classifies plant hydration visually from leaf/root images captured by ESP32-CAM
- A fusion layer combines sensor-based and visual predictions to produce a high-confidence final recommendation
- Mobile app displays live sensor data and ML predictions, allowing farmers to monitor and control irrigation remotely

User Requirements Addressed

- Real-time monitoring of soil moisture, temperature, humidity, and light intensity
- Automated ML-based watering
- Automated ML-based fertilization
- Visual hydration assessment from orchid plant images (CNN model)
- Historical sensor data tracking and moisture trend analysis
- Manual override — farmer can irrigate or fertilize at any time



Design Excellence / Contribution

- Multi-modal ML fusion combining IoT sensor predictions and visual CNN analysis for higher decision reliability
- Three specialized ML models integrated into one pipeline — Random Forest (watering), Decision Tree (fertilization), MobileNetV2 CNN (visual hydration)
- Conflict detection safety mechanism — flags sensor-visual disagreements instead of automating potentially incorrect actions
- Real-time Firebase synchronization enabling remote monitoring and instant prediction updates from any location

User Feedback on Prototype

- Farmers found real-time moisture and watering status display highly practical for daily farmhouse management
- Users requested an auto-watering toggle to maintain manual control — added as a key UI feature based on this feedback
- The visual hydration check using plant images was seen as a valuable secondary verification alongside sensor readings



IT22083500 | Nimla Samadinee L.D

Title of the Individual Component:
Orchid Growth Stage Classification & Bloom Prediction for Vanda Species

Specialization: Software Engineering

The Solution Addresses the Sub-Problem/ Prototype

The prototype uses AI-based image analysis and environmental data to automatically identify Vanda orchid growth stages and predict blooming periods. It helps growers monitor plant development, prepare for flowering, and reduce manual observation errors.

User Requirements Addressed

- Automatic identification of orchid growth stages
- Early prediction of flowering periods
- Monitoring flower blooming and fading stages
- Support for harvest and seed collection planning
- Easy and efficient orchid management for growers



Design Excellence / Contribution

- AI-powered image classification for Vanda orchids
- Multi-modal approach using image and environmental data
- Specialized for Vanda orchid growth behavior
- Improves productivity and decision-making accuracy
- Scalable and user-friendly smart farming solution

User Feedback on Prototype

- Users found the system useful for tracking orchid growth stages
- Bloom prediction feature helps growers plan harvesting and sales
- Reduced the need for continuous manual monitoring
- Users suggested improving prediction accuracy with more training data



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Commercialization / Sustainability of the Solution:

The system can be commercialized as a smart orchid management platform for nurseries and orchid growers. Sustainability is achieved through automated monitoring, efficient water and fertilizer usage, reduced plant losses, and scalable IoT-based deployment.

Customer Persona

- Commercial orchid nursery owner managing 500+ plants
- Greenhouse operator needing automated monitoring
- Orchid exporter wanting better bloom prediction and quality control

Market Size

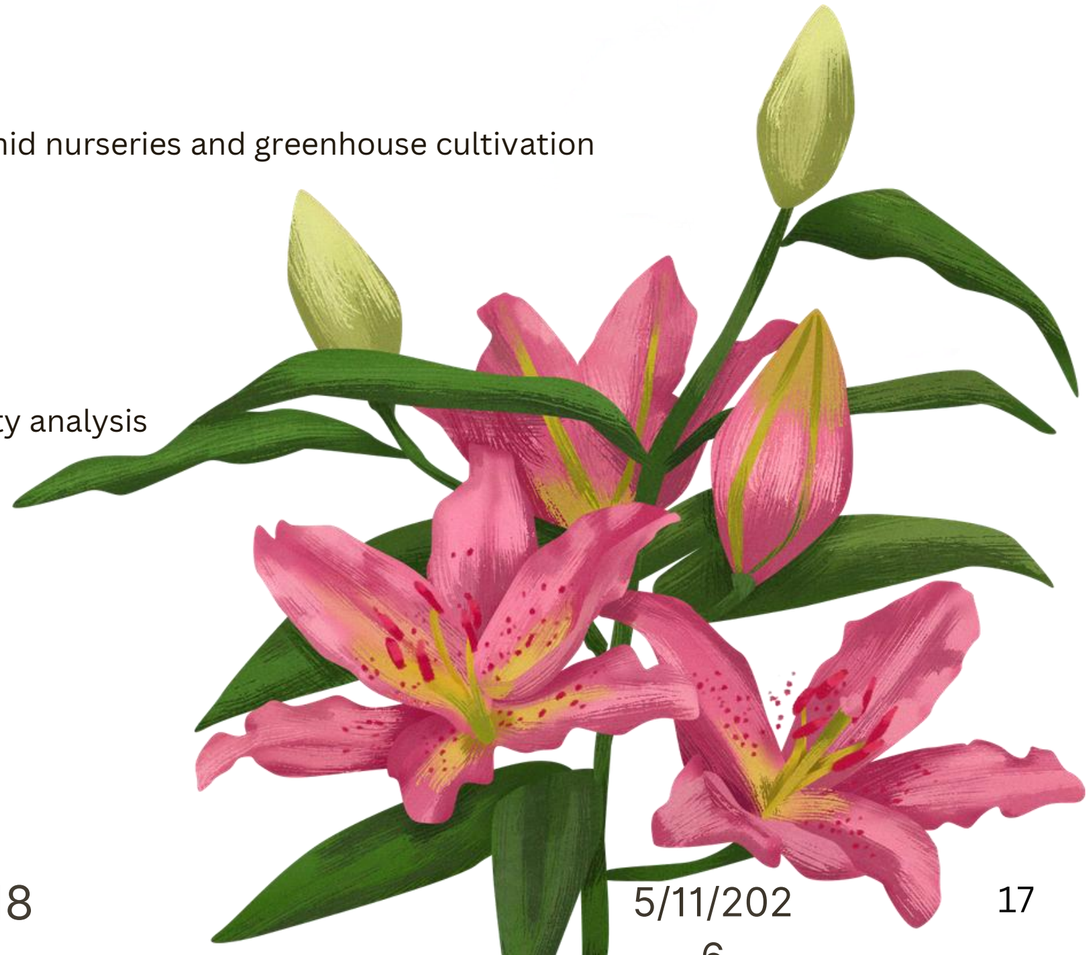
The solution targets the growing floriculture and smart agriculture market, especially orchid nurseries and greenhouse cultivation businesses that require automated plant monitoring and management systems.

What is Unique in the Solution

- Combines AI, IoT, and multi-modal machine learning in one platform
- Specialized for orchid cultivation, especially Vanda species
- Includes disease detection, bloom prediction, smart watering, and hybrid compatibility analysis
- Provides real-time monitoring and predictive decision support
- Cost-effective and scalable for multiple plants and nurseries

How the Cost Will Be Recovered

- Subscription-based software services
- Selling IoT monitoring kits and smart devices
- Licensing the system to nurseries and greenhouse operators
- Maintenance and support services
- Future mobile/web application premium features



Thank you